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Liquid dispenser with plunger

Abstract

A dispenser includes: a plunger; a dispensing portion configured to dispense and suction liquid in response to an advance and a retraction of a distal end surface of the plunger along an axial line. The dispensing portion includes: a surrounding portion surrounding the distal end surface of the plunger about the axial line to define an accommodation chamber for accommodate the liquid to be dispensed; and an end portion facing the distal end surface of the plunger. The surrounding portion includes an inlet opening into the accommodation chamber to receive the liquid into the accommodation chamber. A distance from the inlet to the end portion is less than or equal to half of a maximum stroke of the distal end surface of the plunger.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of PCT Application No. PCT/JP2021/026160, filed on Jul. 12, 2021, which claims the benefit of priority from Japanese Patent Application No. 2020-120737, filed on Jul. 14, 2020, and Japanese Patent Application No. 2020-120741, filed on Jul. 14, 2020. The entire contents of the above listed PCT and priority applications are incorporated herein by reference.

BACKGROUND

Field

(1) The present disclosure relates to a liquid dispenser.

Description of the Related Art

(2) Japanese Unexamined Patent Application Publication No. 2014-163372 discloses a liquid dispense valve that dispenses a predetermined amount of liquid from a tip nozzle by reciprocally driving a plunger from above with respect to the pressurized liquid introduced into a liquid reservoir and a plunger guide. The liquid dispense valve has a connection portion between the plunger and a sliding body of a plunger driver so that a sliding direction of the plunger is not affected even if a sliding direction of the plunger driver is deviated from a sliding direction of the plunger.

SUMMARY

(3) Disclosed herein is A dispenser. The dispenser may include: a plunger comprising a distal end surface facing an advance direction of the plunger and a proximal end surface facing a retraction direction of the plunger; a dispensing portion configured to dispense and suction liquid in response to an advance and a retraction of the distal end surface along an axial line intersecting the distal end surface and the proximal end surface, wherein the dispensing portion comprises: a surrounding portion surrounding the distal end surface of the plunger about the axial line to define an accommodation chamber; and an end portion facing the distal end surface of the plunger along the axial line, wherein the end portion comprises an outlet opening into the accommodation chamber to dispense the liquid out of the accommodation chamber, wherein the surrounding portion comprises an inlet opening into the accommodation chamber to receive the liquid into the accommodation chamber, and wherein a distance from the inlet to the end portion is less than or equal to half of a maximum stroke of the distal end surface of the plunger.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating an example configuration of a dispenser.
- (2) FIG. 2 is a perspective view illustrating an example pump unit.
- (3) FIG. 3 is a cross-sectional view taken along line III-III.
- (4) FIG. 4 is an enlarged view of the vicinity of an example dispensing portion in FIG. 3.
- (5) FIG. 5 is an enlarged view of the vicinity of an example piston in FIG. 3.
- (6) FIG. 6 is an enlarged view of the vicinity of an example advance regulation unit in FIG. 3.
- (7) FIG. 7 is an enlarged view of the vicinity of an example retract regulating unit in FIG. 3.
- (8) FIG. 8 is an enlarged view around an example block assistance unit in FIG. 3.
- (9) FIG. 9 is a schematic view illustrating an example configuration of a connecting portion of a surrounding portion, a cylinder, and an outer cap.
- (10) FIG. 10 is a schematic diagram illustrating an example configuration of a piston driver.
- (11) FIG. 11 is a schematic diagram illustrating an example hardware configuration of control circuitry.

(12) FIG. 12 is a flowchart illustrating an example control procedure by the control circuitry.

DETAILED DESCRIPTION

(13) In the following description, with reference to the drawings, the same reference numbers are assigned to the same components or to similar components having the same function, and overlapping description is omitted.

(14) [Dispenser]

(15) A dispenser **1** shown in FIG. 1 is a device for intermittently dispensing liquid. For example, the dispenser **1** is a jet-type dispenser that intermittently ejects droplets toward a remote application target. Examples of the dispensed liquid include adhesives, lubricants, solder pastes, fluxes, silver pastes, reagents, and the like.

(16) The dispenser **1** has a pump unit **10**, a barrel **20**, and a controller **30**. The pump unit **10** reciprocates a plunger by pressurizing the plunger by gas (hereinafter referred to as “driving gas”). Thus, suction and dispense of the liquid are repeated. The pump unit **10** reciprocates the plunger by switching the supply direction of the driving gas with respect to the plunger by driving power. The barrel **20** pressurizes liquid by gas (hereinafter referred to as “pumping gas”) to supply the liquid to the pump unit **10**. The controller **30** supplies the driving gas and the driving power to the pump unit **10**, and supplies the pumping gas to the barrel **20**. Hereinafter, configurations of the pump unit **10**, the barrel **20**, and the controller **30** will be described in detail.

(17) (Pump Unit)

(18) As shown in FIGS. 2 and 3, the pump unit **10** has a plunger **100**, a guide block **200**, a dispensing portion **300**, a driver **400**, an advance restricting unit **700**, a retraction restricting unit **500**, and a blocking assistance unit **600**.

(19) The plunger **100** is a rod-shaped member formed of a metallic material such as stainless steel, titanium alloy, or cemented carbide, and includes a distal end surface **101** and a proximal end surface **102**. Hereinafter, a direction in which the distal end surface **101** of the plunger **100** faces is referred to as a “advance direction”, and movement in the direction is referred to as “advance”. In addition, a direction in which the proximal end surface **102** of the plunger **100** faces is referred to as a “retraction direction”, and movement in the direction is referred to as a “retraction”.

(20) The plunger **100** has a flange **130** on its outer periphery. The flange **130** has a front surface **131** and a rear surface **132**. The front surface **131** faces the advance direction and the rear surface **132** faces the retraction direction. The flange **130** is positioned closer to the proximal end surface **102** between the distal end surface **101** and the proximal end surface **102**. For example, the distance from the distal end surface **101** to the front surface **131** is longer than the distance from the proximal end surface **102** to the rear surface **132**.

(21) The guide block **200** (guide portion) guides the plunger **100** to advance and retract along an axial line **L1** (for example, a center axial line of the plunger **100**) that intersects (for example, is orthogonal to) the distal end surface **101** and the proximal end surface **102**. The guide block **200** is a cylindrical block member formed of, for example, a fluororesin or an engineering plastic material, and includes a distal end surface **213** and a proximal end surface **214**. The distal end surface **213** faces the advance direction and the proximal end surface **214** faces the retraction direction.

(22) The guide block **200** has a guide hole **215** that penetrates between the distal end surface **213** and the proximal end surface **214** along the axial line **L1**. In the guide hole **215**, the plunger **100** is inserted from the proximal end surface **214** toward the distal end surface **213**, a distal end portion (the distal end surface **101** and its vicinity) of the plunger **100** is located anterior to the distal end surface **213**, and the flange **130** of the plunger **100** is located behind the proximal end surface **214** along the axial line **L1**. By engagement with the guide hole **215**, the plunger **100** is guided to advance and retract along the axial line **L1**.

(23) A distal end fitting portion **211** is formed on an outer peripheral surface of a distal end portion (the distal end surface **213** and its vicinity) of the guide block **200**. A proximal end fitting portion

212 is formed on the outer peripheral surface of a proximal end portion (the proximal end surface **214** and its vicinity) of the guide block **200**.

(24) The dispensing portion **300** accommodates the distal end portion (the distal end surface **101** and its vicinity) of the plunger **100** and dispenses and suctions liquid in response to advance and retraction of the distal end surface **101**. As shown enlarged in FIG. 4, for example, the dispensing portion **300** has a surrounding portion **310**, a nozzle unit **340**, a nozzle seal **351**, a nozzle holder **360**, a plunger seal **352**, and a barrel attachment portion **380**.

(25) The surrounding portion **310** surrounds the distal end surface **101** around the axial line **L1** to define an accommodation chamber **319**. The surrounding portion **310** is a block material formed of a metallic material such as stainless steel or an aluminum alloy, and contains a distal end surface **311** and a proximal end surface **312**. The distal end surface **311** faces the advance direction and the proximal end surface **312** faces the retraction direction.

(26) A circular recess **313** centered on the axial line **L1** is formed on the distal end surface **311**, and a circular recess **314** centered on the axial line **L1** is further formed on the bottom surface of the recess **313**. A circular recess **316** centered on the axial line **L1** is formed on the proximal end surface **312**, and a circular recess **317** centered on the axial line **L1** is further formed on the bottom surface of the recess **316**. The surrounding portion **310** further includes a circular through-hole **321** extending along the axial line **L1** between a bottom surface **315** of the recess **314** and a bottom surface **318** of the recess **317**. The through-hole **321** forms the accommodation chamber **319** and accommodates the distal end portion of the plunger **100**.

(27) The surrounding portion **310** further includes an inlet **323**, a suction port **324**, and an inlet flow channel **325**. The inlet **323** opens into the accommodation chamber **319** and receives the liquid into the accommodation chamber **319**. For example, the inlet **323** opens onto the inner peripheral surface of the through-hole **321**. The suction port **324** opens onto the outer peripheral surface of the surrounding portion **310**.

(28) The inlet flow channel **325** is formed within the surrounding portion **310** to provide communication between the suction port **324** and the inlet **323**. For example, the inlet flow channel **325** is formed by a circular through-hole penetrating between an outer peripheral surface of the surrounding portion **310** and an inner peripheral surface of the through-hole **321**, wherein the opening of the inlet flow channel **325** in the inner peripheral surface of the through-hole **321** forms the inlet **323** and the opening of the inlet flow channel **325** in the outer peripheral surface of the surrounding portion **310** forms the suction port **324**. The inlet flow channel **325** is inclined with respect to a plane vertical to the axial line **L1** so that a portion of the inlet flow channel **325** moves away from the distal end surface **311** as the portion moves away from the axial line **L1**. Therefore, the inlet **323** and the suction port **324** have an elliptical shape extending along the axial line **L1**.

(29) A holder attachment portion **331** is formed on the outer peripheral surface of a distal end portion of the surrounding portion **310** (the distal end surface **311** and its vicinity). The outer periphery of the holder attachment portion **331** is positioned inward from the outer peripheral surface of the surrounding portion **310** on which the suction port **324** is formed. As described above, the inlet flow channel **325** is inclined so that a portion of the inlet flow channel **325** moves away from the distal end surface **311** as the portion moves away from the axial line **L1**, thereby avoiding interference between the outer peripheral surface of the holder attachment portion **331** and the inlet flow channel **325**.

(30) The nozzle unit **340** faces the distal end surface **101** of the plunger **100** along the axial line **L1**. The nozzle unit **340** includes an outlet **343**, a dispense port **344**, and a dispense flow channel **345**. The outlet **343** opens into the accommodation chamber **319** and leads the liquid out of the accommodation chamber **319**. The dispense port **344** opens to an outside of the accommodation chamber **319**. The dispense flow channel **345** connects the outlet **343** and the dispense port **344** without passing through the check valve.

(31) For example, the nozzle unit **340** has a nozzle base **341** and a nozzle **342**. The nozzle base **341**

is a disk-shaped portion formed of, for example, a fluororesin or an engineering plastic material, and is fitted into the recess **314**. The nozzle base **341** includes a through-hole **346** along the axial line **L1**. The nozzle **342** is a narrow tube formed of, for example, stainless steel or an aluminum alloy, and is fixed to the nozzle base **341** in a state of being passed through the through-hole **346**. The lumen of the nozzle **342** forms the dispense flow channel **345**, the opening of the nozzle **342** to the retraction direction forms the outlet **343**, and the opening of the nozzle **342** to the advance direction forms the dispense port **344**.

(32) The nozzle seal **351** provides a seal between the nozzle unit **340** and the surrounding portion **310**. For example, the nozzle seal **351** is an annular seal member (e.g., an O-ring) formed of a rubber material or the like, and is accommodated in the recess **313** in a state of surrounding the nozzle base **341**.

(33) The nozzle holder **360** holds the nozzle unit **340** and is attached to the surrounding portion **310**. For example, the nozzle holder **360** is a member formed of a metallic material such as stainless steel or an aluminum alloy, and includes a cover plate **361** and a peripheral wall **363**. The cover plate **361** covers the distal end surface **311** of the surrounding portion **310**. An opening **362** is formed in the central part of the cover plate **361**. The cover plate **361** holds the nozzle base **341** and the nozzle seal **351** from the advance direction with the nozzle **342** passed through the opening **362**. The peripheral wall **363** rises toward the retraction direction from the periphery of the cover plate **361** and surrounds the holder attachment portion **331**.

(34) The peripheral wall **363** is attached to the holder attachment portion **331**. For example, a male screw **332** is formed on an outer periphery of the holder attachment portion **331**, a female screw **364** corresponding to the male screw **332** is formed on an inner periphery of the peripheral wall **363**, and the peripheral wall **363** is attached to the holder attachment portion **331** by screwing the male screw **332** into the female screw **364**. By screwing the male screw **332** into the female screw **364**, the cover plate **361** approaches the distal end surface **311** and presses the nozzle base **341** and the nozzle seal **351** toward the retraction direction. As the cover plate **361** approaches the distal end surface **311**, the nozzle seal **351** is pressed, thereby strengthening the seal between the nozzle unit **340** and the surrounding portion **310**. Further, the nozzle seal **351** suppresses the deviation between the center of the surrounding portion **310** and the center of the nozzle unit **340**.

(35) The plunger seal **352** faces the nozzle unit **340** across the accommodation chamber **319** and provides a seal between the surrounding portion **310** and the plunger **100**. For example, the plunger seal **352** is an annular seal member formed of a plastic material or the like, and is accommodated in the recess **317** in a state of surrounding the distal end portion of the plunger **100**.

(36) The guide block **200** is accommodated in the recess **316** of the surrounding portion **310**. The distal end fitting portion **211** of the guide block **200** is inserted into the recess **317** and sandwiches the plunger seal **352** with the bottom surface **318** of the recess **317**. This keeps the plunger seal **352** in the recess **317**.

(37) The barrel attachment portion **380** connects the suction port **324** of the surrounding portion **310** and the barrel **20**. For example, the barrel attachment portion **380** has a support arm **381** and an attachment mouthpiece **382** (see FIG. 3). The support arm **381** protrudes outward (toward a direction away from the axial line **L1**) from a portion of the outer periphery of the surrounding portion **310** where the suction port **324** is formed. The attachment mouthpiece **382** protrudes from the end portion of the suction port **324** toward the retraction direction and receives the liquid delivered from the barrel **20**. A relay flow channel **383** is formed in the support arm **381** to communicate the attachment mouthpiece **382** with the suction port **324**.

(38) In the dispensing portion **300** constructed in this manner, the plunger seal **352** provides a seal between the surrounding portion **310** and the plunger **100** to substantially seal the accommodation chamber **319** except for the inlet **323** and the outlet **343**. Therefore, a volume of the accommodation chamber **319** is changed by the advance and retraction of the distal end surface **101** of the plunger **100**.

(39) In response to the advance of the distal end surface **101**, the volume of the accommodation chamber **319** decreases, and the internal pressure of the accommodation chamber **319** increases accordingly. This pressurizes liquid toward both the inlet **323** and the outlet **343**, but since the inlet **323** is pressurized by the barrel **20**, the liquid flows out of the outlet **343** and is dispensed from the dispense port **344**. In response to the retraction of the distal end surface **101**, the volume of the accommodation chamber **319** increases, and the internal pressure of the accommodation chamber **319** decreases accordingly. As a result, liquid flows in from the inlet **323**.

(40) The behavior of the liquid in the dispense flow channel **345** is based on a relationship between the decompression caused by the retraction of the distal end surface **101** and the pressurization caused by the barrel **20** (hereinafter referred to as “pressurization relationship”). For example, if the pressurization relationship is adjusted such that a pressure inside the accommodation chamber **319** is less than or equal to a pressure outside the accommodation chamber **319** (for example, atmospheric pressure) during at least a part of a period during the retraction of the plunger **100** (hereinafter referred to as a “retraction period”), the dispense of liquid from the dispense port **344** ceases during at least a part of the retraction period.

(41) Here, the inlet **323** may be positioned closer to the nozzle unit **340** between the plunger seal **352** and the nozzle unit **340**. The distance from the inlet **323** to the nozzle unit **340** (a distance **D1** in FIG. 4) may be less than or equal to half of a maximum stroke **H1** of the distal end surface **101** of the plunger **100**. The maximum stroke **H1** is a distance from the most retracted position of the end surface **101** to the most advanced position of the distal end surface **101**. More than half of the maximum stroke **H1** may overlap the inlet **323**. For example, an opening height **H2** of the inlet **323** in the direction along the axial line **L1** may be half or more of the maximum stroke **H1**, and the most retracted position of the distal end surface **101** may be located behind the inlet **323**, and the most advanced position of the distal end surface **101** may be located anterior to the inlet **323** along the axial line **L1**. The inlet **323** can be opened at the beginning of the retraction period, so that liquid is suctioned quickly into the accommodation chamber **319**. Also, in most of the retraction period, the opening of the inlet **323** becomes larger as the distal end surface **101** retracts, so that the liquid is suctioned more quickly. Further, since the opening of the inlet **323** becomes smaller as the distal end surface **101** advances in most of a period of the advance of the distal end surface **101**, a pressure into the inlet **323** can be suppressed and the liquid can be rapidly dispensed from the outlet **343**.

(42) The driver **400** causes the driving gas to advance and retract the proximal end portion of the plunger **100**. As shown enlarged in FIG. 5, for example, the driver **400** has a cylinder **410**, a piston **450**, an outer seal **471**, an inner seal **472**, and a piston driver **480**.

(43) The cylinder **410** accommodates the proximal end portion of the plunger **100**. The cylinder **410** is a cylindrical member formed of a metallic material such as stainless steel or an aluminum alloy, and includes a distal end surface **411** and a proximal end surface **412** (see FIG. 3). The distal end surface **411** faces the advance direction and the proximal end surface **412** faces the retraction direction. The cylinder **410** has a second pressurizing hole **432** and a first pressurizing hole **431** arranged in order from the advance direction to the retraction direction. Each of the second pressurizing hole **432** and the first pressurizing hole **431** penetrates between an inner peripheral surface and an outer peripheral surface of the cylinder **410**.

(44) With the cylinder **410** accommodating the proximal end portion of the plunger **100**, the distal end portion (the distal end surface **411** and its vicinity) of the cylinder **410** is connected to the dispensing portion **300**. For example, the distal end portion of the cylinder **410** fits to the proximal end fitting portion **212** of the guide block **200** and is connected to the surrounding portion **310** around the recess **316**.

(45) The piston **450** is provided to the cylinder **410** to partition the cylinder **410** along the axial line **L1** into a first space **413** and a second space **414**. The first space **413** is a space located behind the piston **450** along the axial line **L1**, and the second space **414** is a space located anterior to the piston

450 along the axial line **L1**. The piston **450** causes the plunger **100** to advance in response to the first space **413** pressurized and the plunger **100** to retract in response to the second space **414** pressurized.

(46) The piston **450** is annular and may partition the cylinder **410** into the first space **413** and the second space **414** between the outer peripheral surface of the plunger **100** and the inner peripheral surface of the cylinder **410**. For example, the piston **450** is an annular plate member formed of a metallic material such as stainless steel or an aluminum alloy, and includes a distal end surface **452**, a proximal end surface **453**, and a through-hole **454**. The distal end surface **452** faces the advance direction and the proximal end surface **453** faces the retraction direction. The through-hole **454** penetrates along the axial line **L1** between the distal end surface **452** and the proximal end surface **453**. An inner radius of the through-hole **454** is greater than an outer radius of the proximal end portion of the plunger **100**. On an outer peripheral surface **451** of the piston **450**, a groove **455** is formed over the entire circumference around the axial line **L1**. A recess **456** centered on the axial line **L1** is formed in the distal end surface **452**.

(47) The piston **450** is attached to the proximal end portion of the plunger **100** so that the piston **450** can float (move) in a direction vertical to the axial line **L1**. For example, the driver **400** further comprises a holder **461**. The holder **461** holds the piston **450** around the proximal end portion of the plunger **100** while allowing the piston **450** to float in the direction vertical to the axial line **L1**. For example, the holder **461** sandwiches the piston **450** with the flange **130** with the proximal end portion of the plunger **100** passed through the through-hole **454**.

(48) For example, a holding groove **121** is formed on the outer peripheral surface of the proximal end portion of the plunger **100** over an entire circumference around the axial line **L1**. With the piston **450** in contact with the rear surface **132** of the flange **130**, the holding groove **121** is located behind the piston **450** along the axial line **L1**. The holder **461** is, for example, a C-shaped snap ring, which is fitted into the holding groove **121** that is located behind the piston **450** along the axial line **L1**. Thus, the piston **450** is sandwiched between the flange **130** and the holder **461**. As described above, the inner radius of the through-hole **454** is greater than the outer radius of the proximal end portion of the plunger **100**. The piston **450** is allowed to float in the direction vertical to the axial line **L1** by the difference between the inner radius of the through-hole **454** and the outer radius of the proximal end portion of the plunger **100**.

(49) The outer seal **471** seals between the piston **450** and the cylinder **410**. For example, the outer seal **471** is an annular seal member (e.g., an O-ring) formed of a rubber material or the like, and is accommodated in the groove **455**. The outer seal **471** seals between the piston **450** and the cylinder **410** by contacting the bottom surface of the groove **455** and the inner peripheral surface of the cylinder **410**.

(50) The inner seal **472** (seal portion) seals between the piston **450** and the plunger **100** while allowing the piston **450** to float in the direction vertical to the axial line **L1**. For example, the inner seal **472** is an annular seal member (for example, an O-ring) formed of a rubber material or the like, and is accommodated in the recess **456** in a state of surrounding the plunger **100**. The inner seal **472** seals between the piston **450** and the plunger **100** by contacting a bottom surface **457** (annular first seal surface) of the recess and the rear surface **132** (annular second seal surface facing the first seal surface) of the flange **130**.

(51) Since the bottom surface **457** and the rear surface **132** intersect (for example, are orthogonal to) the axial line **L1**, even if the piston **450** floats in a vertical direction in the axial line **L1**, the inner seal **472** is kept in contact with the bottom surface **457** and the rear surface **132**. Therefore, both the float property of the piston **450** and the seal property between the piston **450** and the plunger **100** can be achieved.

(52) The piston driver **480** switches between a first state in which a pressure from a pressurizing source (a pressure of the above mentioned driving gas) is applied to the first space **413** and a second state in which the pressure from the pressurizing source is applied to the second space **414**

in accordance with the supply of the driving power. For example, the piston driver **480** may pressurize the first space **413** when there is no supply of the driving power and pressurize the second space **414** when there is the supply of the driving power. An example structure of the piston driver **480** will be described later.

(53) The advance restricting unit **700** regulates an advance of the plunger **100**. For example, the advance restricting unit **700** is provided between the guide block **200** and the flange **130**, and regulates an advance of the flange **130**. For example, the advance restricting unit **700** may be placed in the second space **414** to provide a seal between the inner peripheral surface of the cylinder **410** and the outer peripheral surface of the plunger **100** while restricting the advance of the flange **130**. As shown enlarged in FIG. 6, by way of example, the advance restricting unit **700** has an advance restricting block **710**, outer seals **721**, **722**, and inner seals **731**, **732**. The advance restricting block **710** (advance regulating portion) is a cylindrical block member formed of a metallic material such as stainless steel or an aluminum alloy, and includes a distal end surface **711** and a proximal end surface **712**. The distal end surface **711** faces the advance direction and the proximal end surface **712** faces the retraction direction. The advance restricting block **710** has a guide hole **713** that penetrates between the distal end surface **711** and the proximal end surface **712** along the axial line L1. The plunger **100** is inserted into the guide hole **713** from the proximal end surface **712** toward the distal end surface **711**. The proximal end surface **712** of the advance restricting block **710** faces the front surface **131** of the flange **130** to regulate the advance of the flange **130**.

(54) A groove **714** is formed on an outer peripheral surface of a distal end portion (the distal end surface **711** and its vicinity) of the advance restricting block **710** over an entire circumference around the axial line L1. A groove **715** is formed on an outer peripheral surface of a proximal end portion (the proximal end surface **712** and its vicinity) of the advance restricting block **710** over an entire circumference around the axial line L1. Grooves **716**, **717** arranged along the axial line L1 are formed on an inner peripheral surface of the guide hole **713**. Each of the grooves **716**, **717** extends around an entire circumference of the axial line L1.

(55) Each of the outer seals **721**, **722** is an annular seal member (for example, an O-ring) formed of a rubber material or the like and seals between the advance restricting block **710** and the cylinder **410**. For example, an outer seal **721** may be accommodated in the groove **714** in a state of surrounding the advance restricting block **710**, and may be in contact with a bottom surface of the groove **714** and the inner peripheral surface of the cylinder **410** to seal between the advance restricting block **710** and the cylinder **410**. An outer seal **722** may be accommodated in the groove **715** in a state of surrounding the advance restricting block **710**, and may be in contact with the bottom surface of the groove **715** and the inner peripheral surface of the cylinder **410** to seal between the advance restricting block **710** and the cylinder **410**.

(56) Each of the inner seals **731**, **732** is an annular seal member (for example, an O-ring) formed of a rubber material or the like, and seals between the advance restricting block **710** and the plunger **100**. For example, an inner seal **731** may be accommodated in the groove **716** in a state of surrounding the plunger **100**, and may be in contact with the bottom surface of the groove **716** and the outer peripheral surface of the plunger **100** to seal between the advance restricting block **710** and the plunger **100**. An inner seal **732** may be accommodated in the groove **717** in a state of surrounding the plunger **100**, and may be in contact with the bottom surface of the groove **717** and the outer peripheral surface of the plunger **100** to seal between the advance restricting block **710** and the plunger **100**.

(57) The pump unit **10** further comprises an advance limit adjuster **420**. The advance limit adjuster **420** adjusts the position of the advance restricting unit **700** relative to the cylinder **410** in a direction along the axial line L1. For example, the advance limit adjuster **420** has a male screw **718**, a female screw **421**, and an adjustment window **422**. The male screw **718** is formed around the outer peripheral surface of the advance restricting block **710** between the groove **714** and the

groove **715**. The female screw **421** is formed on a portion of the inner peripheral surface of the cylinder **410** corresponding to the male screw **718**. The advance restricting unit **700** is placed in the cylinder **410** with the male screw **718** screwed into the female screw **421**.

(58) The adjustment window **422** (see FIG. 2) penetrates between the inner peripheral surface and the outer peripheral surface of the cylinder **410** at a position located anterior to the groove **714** along the axial line **L1**. The adjustment window **422** exposes a portion of the outer peripheral surface of the advance restricting block **710** to the exterior of the cylinder **410**. Thus, an operation force around the axial line **L1** can be applied to the outer peripheral surface of the advance restricting block **710** from the outside of the cylinder **410**, and the advance restricting block **710** can be rotated around the axial line **L1**.

(59) Rotating the advance restricting block **710** rotates the male screw **718** relative to the female screw **421** and displaces the advance restricting block **710** along the axial line **L1**. This adjusts the position of the advance restricting unit **700** relative to the cylinder **410**. The above-described maximum stroke **H1** is a stroke in a state in which the advance restricting unit **700** is located at the most advanced position along the axial line **L1**.

(60) The retraction restricting unit **500** regulates the retraction of the plunger **100**. For example, the retraction restricting unit **500** is connected to the proximal end portion of the cylinder **410** and regulates the retraction of the proximal end surface **102** of the plunger **100**. As shown enlarged in FIG. 7, for example, the retraction restricting unit **500** has an outer cap **510**, a regulating rod **520**, a retraction limit adjustment portion **530**, an inner cap **540**, an outer seal **551**, and an inner seal **552**.

(61) The outer cap **510** is a plate-shaped member formed of a metallic material such as stainless steel or an aluminum alloy, and closes the proximal end portion of the cylinder **410**. The outer cap **510** has an opening **511** in its center.

(62) The regulating rod **520** is a rod-shaped member formed of a metallic material such as stainless steel or an aluminum alloy, and is inserted into the opening **511** along the axial line **L1**. The regulating rod **520** has a distal end surface **521** and a proximal end surface **522**. The distal end surface **521** faces the advance direction and the proximal end surface **522** faces the retraction direction. The distal end surface **521** (retraction restricting portion) faces the proximal end surface **102** of the plunger **100** in the cylinder **410** and restricts the retraction of the plunger **100**. The proximal end surface **522** is positioned outside the cylinder **410**.

(63) The retraction limit adjustment portion **530** accommodates the regulating rod **520** outside the cylinder **410**. The retraction limit adjustment portion **530** has a handle **531** and causes the regulating rod **520** to advance and retract in response to the rotation of the handle **531** about the axial line **L1**. This adjusts the position of the distal end surface **521** in a direction along the axial line **L1**. The above-described maximum stroke **H1** is a stroke in a state in which the distal end surface **521** is located at the most retracted position along the axial line **L1**.

(64) The inner cap **540**, the outer seal **551** and the inner seal **552** provide a seal between the cylinder **410** and the regulating rod **520** in the cylinder **410**. For example, the inner cap **540** has a through-hole **542** in its center. The inner cap **540** is placed in the cylinder **410** and secured to the outer cap **510** with the regulating rod **520** passed through the through-hole **542**.

(65) The outer seal **551** seals between the inner cap **540** and the cylinder **410** by contacting the outer peripheral surface of the inner cap **540** and the inner peripheral surface of the cylinder **410**. A flange **543** is formed on an outer peripheral surface of the inner cap **540** and the outer seal **551** is held between the flange **543** and the outer cap **510**.

(66) The inner seal **552** seals between the inner cap **540** and the regulating rod **520** by contacting an inner peripheral surface of the through-hole **542** and an outer peripheral surface of the regulating rod **520**. An inward flange **544** is formed on the inner peripheral surface of the through-hole **542** and the inner seal **552** is held between the inward flange **544** and the outer cap **510**.

(67) When the pressurizing source is not supplied to the piston driver **480**, the blocking assistance unit **600** applies a repulsive force to the piston **450** toward the advance direction to keep the

plunger **100** at the most advanced position (the position in which the front surface **131** is in contact with the advance restricting block **710**). This keeps the distal end surface **101** of the plunger **100** in close proximity to the nozzle unit **340**, thereby keeping the outlet **343** substantially blocked and preventing leakage of liquid from the outlet **343**.

(68) For example, the blocking assistance unit **600** has a spring **620** and a pusher **630**. For example, the spring **620** applies a repulsive force to the plunger **100** toward the advance direction. For example, the spring **620** is a coil spring that surrounds the regulating rod **520** and generates a repulsive force against compression along the axial line **L1** direction. The pusher **630** intervenes between the spring **620** and the piston **450** and transmits the repulsive forces generated by the spring **620** to the piston **450**. In this configuration, the spring **620** applies the repulsive force to the piston **450** between the outer peripheral surface of the plunger **100** and the inner peripheral surface of the cylinder **410** via the pusher **630**.

(69) As shown enlarged in FIG. **8**, the pusher **630** is a cylindrical block member formed of a metallic material such as stainless steel or aluminum alloy, and is disposed in the cylinder **410** along the axial line **L1**. The pusher **630** has a distal end surface **641** and a proximal end surface **642**. The distal end surface **641** faces the advance direction and the proximal end surface **642** faces the retraction direction. A recess **634** centered on the axial line **L1** is formed in the proximal end surface **642**. The spring **620** is accommodated in the recess **634**. The spring **620** exerts a repulsive force on the bottom surface of the recess **634**.

(70) A recess **635** centered on the axial line **L1** is formed in the distal end surface **641**, and a recess **636** centered on the axial line **L1** is further formed in the bottom surface of the recess **635**. An opening **637** centered on the axial line **L1** is formed between the bottom surface of the recess **636** and the bottom surface of the recess **634**, and the regulating rod **520** is passed through the opening **637**. A plurality of vent holes **638** are formed between the bottom surface of the recess **635** and the bottom surface of the recess **634**. Thus, the space in which the piston **450** is disposed and the space in which the spring **620** is disposed communicate with each other. When the pusher **630** contacts the piston **450**, the recess **635** receives the holder **461** and the recess **636** receives the proximal end portion of the plunger **100**.

(71) The pump unit **10** further comprises a release portion **440**. The release portion **440** releases the application of a repulsive force to the piston **450** by the spring **620** by the pressure of the pressurizing source (for example, the pressure of the driving gas). Therefore, when the pressure of the pressurizing source is supplied, the repulsive force of the spring **620** does not act as resistance against the sliding of the piston **450**, and the piston **450** can slide at high speed. For example, the release portion **440** forms a third space, between the pusher **630** and the cylinder **410**, to which the pressure of the pressurizing source is applied in both the first state and the second state, and releases the application of the repulsive force by the spring **620** by pressurizing of the third space. As an example, the release portion **440** has switching seals **651**, **652**, and a third pressurizing hole **443**.

(72) In the cylinder **410**, an inner diameter of a portion that accommodates a distal end portion (the distal end surface **641** and its vicinity) of the pusher **630** (hereinafter referred to as “a first accommodation portion **441**”), and an inner diameter of a portion that accommodates a proximal end portion (the proximal end surface **642** and its vicinity) of the pusher **630** (hereinafter referred to as “a second accommodation portion **442**”) are different with each other. In particular, the inner diameter of the second accommodation portion **442** is greater than the inner diameter of the first accommodation portion **441**. On the inner peripheral surface of the first accommodation portion **441**, a groove **444** is formed over an entire circumference around an axial line **L1**. A flange **631** is formed on the outer peripheral surface of the proximal end portion of a pusher **630**. On the outer peripheral surface of the flange **631**, a groove **633** is formed over an entire circumference around the axial line **L1**.

(73) A switching seal **651** provides a seal between the pusher **630** and the first accommodation

portion **441**. For example, the switching seal **651** may be an annular seal member (e.g., an O-ring) formed of a rubber material or the like and accommodated in the groove **444** surrounding the distal end portion of the pusher **630**. The switching seal **651** seals between the pusher **630** and the first accommodation portion **441** by contacting the bottom surface of the groove **444** and the outer peripheral surface of the pusher **630**.

(74) A switching seal **652** provides a seal between the pusher **630** and the second accommodation portion **442**. For example, the switching seal **652** is an annular seal member (for example, an O-ring) formed of a rubber material or the like, and is accommodated in the groove **633** surrounding the flange **631** of the pusher **630**. The switching seal **652** seals between the pusher **630** and the second accommodation portion **442** by contacting the bottom surface of the groove **633** and the inner peripheral surface of the second accommodation portion **442**. The third pressurizing hole **443** penetrates between the inner peripheral surface and the outer peripheral surface of the cylinder **410** between the switching seal **651** and the switching seal **652**. The third pressurizing hole **443** is connected to the pressurizing source in both the first state and the second state.

(75) With the above configuration, the third space **445** is formed between the switching seal **651** and the switching seal **652**, which is sealed except for the third pressurizing hole **443** and is connected to the pressurizing source in both of the first state and the second state. As the third pressurizing hole **443** is pressurized by the pressurizing source, the pusher **630** retracts with the switching seal **652** to correspondingly enlarge the third space **445**. This moves the pusher **630** away from a piston **450** and releases the application of the repulsive force by the spring **620** to the piston **450**. As described above, since the space in which the piston **450** is disposed and the space in which the spring **620** is disposed communicate with each other by the plurality of vent holes **638** formed in the pusher **630**, a difference in pressures between these two spaces is less likely to occur. Therefore, the pusher **630** smoothly rebounds in response to the pressurizing of the third space **445**.

(76) The above-described plunger **100**, guide block **200**, dispensing portion **300**, driver **400**, advance restricting unit **700**, retraction restricting unit **500**, and blocking assistance unit **600** are integrated by connecting the surrounding portion **310**, the cylinder **410**, and the outer cap **510**.

(77) As shown in FIG. **9**, the surrounding portion **310**, the cylinder **410**, and the outer cap **510** may be connected by a plurality of through bolts **11** (fastening members) that are inserted into the outer cap **510** from the retraction direction toward the advance direction and reach the surrounding portion **310**. Since the removal of the through bolts **11** can be performed from the opposite side of the nozzle unit **340** dispensing the liquid, the maintenance workability is improved.

(78) Here, the configuration of the above piston driver **480** will be described. As shown in FIG. **10**, the piston driver **480** has a pressurizing port **481**, exhaust ports **482**, **483**, a first flow channel **484**, a second flow channel **485**, a third flow channel **486**, a solenoid valve **487**, an elastic member **488**, and a solenoid **489**.

(79) The pressurizing port **481** is a port for supplying the driving gas. Exhaust ports **482**, **483** are ports for exhausting gas in the cylinder **410**. The first flow channel **484** is a flow channel connected to the first pressurizing hole **431**, and the second flow channel **485** is a flow channel connected to the second pressurizing hole **432**. The third flow channel **486** is a flow channel connecting the pressurizing port **481** and the third pressurizing hole **443**.

(80) The solenoid valve **487** moves between a first position and a second position. The solenoid valve **487** connects the first flow channel **484** and the pressurizing port **481**, and connects the second flow channel **485** and the exhaust port **482** in the first position. Hereinafter, this state is referred to as a first state. In the first state, the first space **413** is pressurized by the driving gas, gas of the second space **414** is exhausted from the exhaust port **482**, and the piston **450** advances.

(81) The solenoid valve **487** connects the second flow channel **485** and the pressurizing port **481**, and connects the first flow channel **484** and the exhaust port **483** in the second position.

Hereinafter, this state is referred to as a second state. In the second state, the second space **414** is pressurized by the driving gas, gas of the first space **413** is exhausted from the exhaust port **483**,

and the piston **450** retracts.

(82) The elastic member **488** applies an elastic repulsive force from the second position toward the first position to the solenoid valve **487**. The solenoid **489** applies a driving force from the first position toward the second position to the solenoid valve **487** by the supplied driving power. Therefore, in a state in which the driving power is not supplied to the solenoid **489**, the solenoid valve **487** is disposed at the first position by the elastic repulsive force of the elastic member **488**. In a state where the driving power is supplied to the solenoid **489**, the solenoid valve **487** is arranged at the second position by the driving force against the elastic repulsive force.

(83) According to such a configuration, even in a case where the driving power is not supplied due to a failure of a controller **30** described later or the like, the plunger **100** is maintained in the most advanced state as long as the supply of the driving gas is continued. This prevents leakage of liquid since the outlet **343** is substantially blocked.

(84) The third pressurizing hole **443** is directly connected to the pressurizing port **481** by the third flow channel **486**. Therefore, in both the first state and the second state, the third space **445** is pressurized by the driving gas. Therefore, as long as the driving gas is supplied to the pressurizing port **481**, the application of the repulsive force to the piston **450** by the spring **620** is released, and the piston **450** can readily be operated at high speed.

(85) When the supply of the driving gas to the pressurizing port **481** is stopped due to a failure of the controller **30** or the like, since the driving gas is not supplied to the third space **445**, the application of the repulsive force to the piston **450** by the spring **620** is resumed and the plunger **100** is kept in the most advanced state. This prevents leakage of liquid since the outlet **343** is substantially blocked.

(86) (Barrel)

(87) The barrel **20** pumps liquid from outside the accommodation chamber **319** to the inlet **323**. As shown in FIG. **1**, for example, the barrel **20** has a delivery port **21** and a pressurizing port **22**. The delivery port **21** is attached to the attachment mouthpiece **382** of the barrel attachment portion **380** and pumps liquid through the attachment mouthpiece **382** to the relay flow channel **383**. The pressurizing port **22** receives the pumping gas described above. The barrel **20** pressurizes the liquid with pumping gas coming from the pressurizing port **22** and pumps the liquid through the delivery port **21**, the attachment mouthpiece **382**, the relay flow channel **383**, the suction port **324**, and the inlet flow channel **325** into the inlet **323**.

(88) (Control Device)

(89) As shown in FIG. **10**, the controller **30** has air circuitry **40** and control circuitry **50**. The air circuitry **40** supplies the driving gas to the pump unit **10** and the pumping gas to the barrel **20**. For example, the air circuitry **40** is connected to a gas source via an input hose **73**, to the pressurizing port **481** in the pump unit **10** via an output hose **71**, and to the pressurizing port **22** of the barrel **20** via an output hose **72**.

(90) The air circuitry **40** outputs a portion of gas (hereinafter referred to as “input gas”) that flows in from the input hose **73**, and output remaining portion of the input gas to the output hose **72** as the pumping gas. The air circuitry **40** has a regulator **41**, an electropneumatic regulator **42**, a valve **43**, and pressure sensors **44**, **45**, **46**.

(91) The regulator **41** reduces pressure of the input gas to a supply pressure of the driving gas and outputs it to the output hose **71** and the electropneumatic regulator **42**. The electropneumatic regulator **42** further reduces the pressure reduced by the regulator **41** to a supply pressure of the pumping gas and outputs it to the output hose **72**.

(92) The electropneumatic regulator **42** changes the supply pressure of the pumping gas according to a control command. The valve **43** is, for example, a solenoid valve, and opens and closes a flow path between the electropneumatic regulator **42** and the output hose **72** according to a control command.

(93) The pressure sensor **44** detects the pressure of the input gas prior to passing through the

electropneumatic regulator **42**. The pressure sensor **45** detects the pressure of the driving gas between the electropneumatic regulator **42** and the output hose **71**. The pressure sensor **46** detects the pressure of the pumping gas between the electropneumatic regulator **42** and the valve **43**.

(94) The control circuitry **50** controls the air circuitry **40**. The control circuitry **50** is also connected to the piston driver **480** via a cable **74** and controls the piston driver **480**. For example, the control circuitry **50** has a pressure monitoring unit **51**, a barrel pressure control unit **52**, and a dispense control unit **53** as a functional elements (hereinafter referred to as a “functional block”).

(95) The pressure monitoring unit **51** adjusts the pressures applied by the barrel **20** to the liquid. For example, the pressure monitoring unit **51** switches between supplying and stopping the pumping gas by opening and closing the valve **43**. The barrel pressure control unit **52** controls the electropneumatic regulator **42** to cause the supply pressure of the pumping gas to follow a target pressure. The target pressure is determined based on, for example, a setting input of a user. The setting input is acquired by an input device **66** that is described later. The dispense control unit **53** supplies the driving power to the piston driver **480** so as to repeat switching between the first state and the second state at a predetermined cycle at a predetermined period. For example, the dispense control unit **53** supplies the driving power to the piston driver **480** according to a dispense command from a host controller **80**. The dispense command includes, for example, the predetermined cycle and the predetermined period.

(96) FIG. **11** is a block diagram illustrating the hardware configuration of the control circuitry **50**. As shown in FIG. **11**, the control circuitry **50** includes a processor **61**, a memory **62**, a storage **63**, an input/output port **64**, a display device **65**, the input device **66**, and a communication port **67**. Although one processor **61** is shown in the figure, the control circuitry **50** may have a plurality of processors **61**. The control circuitry **50** may have the memory **62** and the storage **63** for each the processor **61**.

(97) The storage **63** may include one or more storage devices each of which include one or more non-transitory and computer-readable storage mediums, such as a nonvolatile semiconductor memory. The storage **63** stores a program for causing the control circuitry **50** to configure the functional blocks described above. The memory **62** may include one or more memory devices each of which temporarily stores the program loaded from the storage medium of the storage **63** and the calculation result by the processor **61**. The one or more memory devices are, for example, a random access memory. The processor **61** may include one or more processing devices and configures the functional blocks of the control circuitry **50** by executing the program in cooperation with the memory **62**. The input/output port **64** inputs and outputs electrical signals to and from the valve **43**, pressure sensors **44**, **45**, **46** and the solenoid valve **487** in accordance with instructions from the processor **61**. The display device **65** includes, for example, a liquid crystal panel or an organic EL panel, and displays an interface image in accordance with instructions from the processor **61**. The input device **66** includes, for example, an input key and acquires an input (key input) to the input key. The display device **65** and the input device **66** may be integrated as a touch panel **33** (see FIG. **1**). The communication port **67** performs information communication with the host controller **80** in accordance with instructions from the processor.

(98) The control circuitry **50** may not be limited to one in which each function is configured by a program. For example, at least a part of the functions of the control circuitry **50** may be configured by a dedicated logic circuit or an application specific integrated circuit (ASIC) in which the dedicated logic circuit is integrated.

(99) [Control Procedure]

(100) Hereinafter, a control procedure by the control circuitry **50** will be described. As shown in FIG. **12**, the control circuitry **50** first executes operations **S01** and **S02**. In the operation **S01**, the pressure monitoring unit **51** acquires the detection results of the pressures by the pressure sensors **44**, **45**, **46**. In the operation **S02**, the pressure monitoring unit **51** checks whether the pressures detected by the pressure sensors **44**, **45**, **46** are within a normal range.

(101) When it is determined that at least one of the detection results of the pressures in the operation **S02** is not within the normal range, the control circuitry **50** performs a operation **S21**. In the operation **S21**, the pressure monitoring unit **51** notifies the host controller **80** of the error. The control circuitry **50** then terminates the control procedure.

(102) When it is determined that the detection results of the pressures in the operation **S02** is within the normal range, the control circuitry **50** executes operations **S03**, **S04**. In the operation **S03**, the barrel pressure control unit **52** opens the valve **43** to start the supply of pumping gas to the barrel **20**. In the operation **S04**, the dispense control unit **53** checks whether or not there is a dispense command from the host controller **80** or the like.

(103) When it is determined that there is the dispense command in the operation **S04**, the control circuitry **50** executes a operation **S05**. In the operation **S05**, the dispense control unit **53** supplies the driving power to the piston driver **480** so as to repeat dispense and suction of the liquid according to the dispense command.

(104) Next, the control circuitry **50** performs operations **S06**, **S07**. If it is determined that there is no dispense command in the operation **S04**, the control circuitry **50** executes the operations **S06**, **S07** without executing the operation **S05**. In the operation **S06**, the pressure monitoring unit **51** acquires the detection results of the pressures by the pressure sensors **44**, **45**, **46**. In the operation **S07**, the pressure monitoring unit **51** checks whether the pressures detected by the pressure sensors **44**, **45**, **46** are within the normal range.

(105) When it is determined that the detection results of the pressures in the operation **S07** is within the normal range, the control circuitry **50** performs a operation **S08**. In the operation **S08**, the barrel pressure control unit **52** checks whether a control stop command is received from the host controller **80**.

(106) When it is determined that the control stop command is not received in the operation **S08**, the control circuitry **50** returns the processing to the operation **S04**. Thereafter, until an abnormality occurs in the detection results of the pressure sensors **44**, **45**, **46** or the control stop command is received from the host controller **80**, the control of dispensing the liquid to the pump unit **10** according to the dispense command is repeated.

(107) When it is determined that at least one of the detection results of the pressures in the operation **S07** is not within the normal range, the control circuitry **50** executes a operation **S11**. In the operation **S11**, the pressure monitoring unit **51** notifies the host controller **80** of the error.

(108) If it is determined that the control stop command is received in the operation **S08**, or after the operation **S11**, the control circuitry **50** executes a operation **S12**. In the operation **S12**, the barrel pressure control unit **52** closes the valve **43** and stops the supply of the pumping gas to the barrel **20**. The control circuitry **50** then terminates the control procedure.

(109) As described above, the dispenser **1** includes: a plunger **100** comprising a distal end surface **101** facing an advance direction and a proximal end surface **102** facing a retraction direction; a guide block **200** configured to guide an advance toward the advance direction and retraction toward the retraction direction of the plunger **100** along an axial line **L1** intersecting the distal end surface **101** and the proximal; a dispensing portion **300** configured to dispense and suction liquid in response to the advance and the retraction of the distal end surface **101**; a driver **400** configured to cause the plunger **100** to advance toward the advance direction and retract toward the retraction direction. The dispensing portion **300** includes: a surrounding portion **310** surrounding the distal end surface **101** of the plunger **100** about the axial line **L1** to define an accommodation chamber **319**; a nozzle unit **340** facing the distal end surface **101** of the plunger **100** along the axial line **L1**; and a plunger seal **352** facing the nozzle unit **340** so that the accommodation chamber **319** is located between the plunger seal **352** and the nozzle unit **340**, and sealing between the surrounding portion **310** and the plunger **100**. The nozzle unit **340** includes an outlet **343** opening into the accommodation chamber **319** to dispense the liquid out of the accommodation chamber **319**. The surrounding portion **310** includes an inlet **323** opening into the accommodation chamber **319** to

receive the liquid into the accommodation chamber **319**. The inlet **323** is located closer to the nozzle unit **340** between the plunger seal **352** and the nozzle unit **340**.

(110) According to this dispenser **1**, the inlet **323** is opened at the beginning of a period during which the plunger retracts (hereinafter referred to as a “retraction period”), so that liquid is rapidly suctioned into the accommodation chamber **319**. In addition, since the opening areas of the inlet **323** are increased as the plunger **100** retracts, the liquid can be suctioned more quickly. Further, the opening areas of the inlet **323** become smaller as the plunger **100** advances in a period during which the plunger **100** advance (hereinafter referred to as “advance period”), the pressure from escaping to the inlet **323** can be suppressed and the liquid can be dispensed more quickly from the outlet **343**. Therefore, dispensing may be sped up.

(111) A distance from the inlet **323** to the nozzle unit **340** is less than or equal to half of a maximum stroke **H1** of the distal end surface **101** of the plunger **100**. Liquid can be suctioned more quickly.

(112) More than half of the maximum stroke **H1** of the distal end surface **101** of the plunger **100** may overlap the inlet **323** along the axial line **L1**. Liquid can be suctioned more quickly.

(113) The nozzle unit **340** may further include: a dispense port **344** opening to outside of the accommodation chamber **319**; and a dispense flow channel **345** connecting the outlet **343** and the dispense port **344** without passing through a check valve. Since there is no resistance of the check valve, the liquid can be dispensed more quickly. In addition, since the inlet **323** opens early when sucking the liquid, the liquid can be suppressed from being drawn into the outlet **343** even without the check valve.

(114) The driver **400** may include: a cylinder **410** accommodating the plunger **100**; a piston **450** partitioning an inside of the cylinder **410** into a first space **413** and a second space **414**, configured to: advance the plunger **100** toward the advance direction according to a pressure of the first space **413**; and retract the plunger **100** toward the retraction direction according to a pressure of the second space **414**; and a piston driver **480** configured to switch between a first state in which a pressure of a pressurizing source is applied to the first space **413** and a second state in which the pressure of the pressurizing source is applied to the second space **414**. Even when the driving power is not supplied due to a failure of the control system, the cylinder **410** can be held at the most advanced position as long as the pressurizing is continued, and leakage of the liquid from the outlet **343** can be suppressed. Therefore, both of an increase in dispensing speed and an improvement in reliability can be achieved.

(115) The piston driver **480** may pressurize the first space **413** or the second space **414** by a pressure of a pressing source. The dispenser **1** may further include a spring **620** configured to apply a repulsive force to the plunger **100** toward the advance direction; and a release portion **440** configured to release, by the pressure of the pressurizing source, an application of the repulsive force from the spring **620** to the plunger **100**. When the pressurizing source is supplied, the repulsive force applied to the piston **450** by the spring **620** may be released to allow the piston **450** to slide at high speed, and when the pressurizing source is not supplied, the plunger **100** may be maintained at the most advanced position by the repulsive force of the spring **620**. Accordingly, leakage of the liquid from the outlet **343** can be more reliably suppressed.

(116) The dispenser **1** may further comprise a barrel **20** configured to pump the liquid from outside the accommodation chamber **319** to the inlet **323**; and a pressure monitoring unit **51** configured to adjust a pressure applied to the liquid by the barrel **20**. The dispensing speed can be increased by optimizing the relationship between the pressurizing and depressurizing by the advance and retract of the plunger **100** and the pressurizing by the barrel **20**.

(117) As described above, the dispenser **1** includes: a plunger **100** comprising a distal end surface **101** facing an advance direction and a proximal end surface **102** facing a retraction direction; a guide block **200** configured to guide an advance toward the advance direction and retraction toward the retraction direction of the plunger **100** along an axial line **L1** intersecting the distal end surface **101** and the proximal; a dispensing portion **300** configured to dispense and suction liquid in

response to the advance and the retraction of the distal end surface **101**; a cylinder **410** accommodating the plunger **100**; a piston **450** partitioning an inside of the cylinder **410** into a first space **413** and a second space **414**, configured to: advance the plunger **100** toward the advance direction according to a pressure of the first space **413**; and retract the plunger **100** toward the retraction direction according to a pressure of the second space **414**; a piston driver **480** configured to switch between a first state in which a pressure of a pressurizing source is applied to the first space **413** and a second state in which the pressure of the pressurizing source is applied to the second space **414**; a spring **620** configured to apply a repulsive force to the plunger **100** toward the advance direction; and a release portion **440** configured to release, by the pressure of the pressurizing source, an application of the repulsive force from the spring **620** to the plunger **100**. (118) According to this dispenser **1**, when the pressurizing source is supplied, the repulsive force applied to the piston **450** by the spring **620** is released to enable high-speed sliding in the piston **450**, and when the pressurizing source is not supplied, the plunger **100** can be held at the most advanced position by the repulsive force of the spring **620**. Therefore, both high speed dispensing and reliability can be achieved.

(119) The release portion **440** may form a third space **445** in the cylinder **410**, so that the pressure of the pressurizing source is applied to the third space **445** in both the first state and the second state. The application of the repulsive force may be released by the pressure of the third space **445**. The configuration of the release portion **440** can be simplified.

(120) The piston **450** may have an annular shape and partitions the inside of the cylinder **410** into the first space **413** and the second space **414** between an outer peripheral surface of the plunger **100** and an inner peripheral surface of the cylinder **410**. The spring **620** may apply the repulsive force to the piston **450** between the outer peripheral surface of the plunger **100** and the inner peripheral surface of the cylinder **410**. By utilizing the space around the proximal end portion of the plunger **100** as the arrangement space of the spring **620** and the release portion **440**, increase in size due to the addition of the spring **620** and the release portion **440** can be suppressed.

(121) The dispenser **1** may further comprise a holder **461** holding the piston **450** on the outer peripheral surface of the plunger **100** while allowing the piston **450** to move in a direction vertical to the axial line L1. By the holder **461** that allows the axial line L1 to float the piston **450** in the vertical direction, both the position of the plunger **100** with respect to the guide block **200** and the position of the piston **450** with respect to the cylinder **410** can be optimized, and the speed of sliding of the piston **450** and the plunger **100** can be increased.

(122) The driver **400** may further have an annular inner seal **472** sealing between the piston **450** and the plunger **100** while allowing the piston **450** to move in the direction vertical to the axial line L1. By sealing the piston **450** formed by the annular shape over the entire circumference, the first space **413** and the second space **414** may be firmly pressurized and the piston **450** may be driven at a higher speed.

(123) The piston **450** may have an annular bottom surface **457** along a plane intersecting the axial line L1. The plunger **100** may have an annular rear surface **132** facing the bottom surface **457**. The plunger seal **352** may contact the bottom surface **457** and the rear surface **132** to seal between the piston **450** and the plunger **100**. By interposing the inner seal **472** between the bottom surface **457** and the rear surface **132**, both the float property in the direction vertical to the axial line L1 and the seal property of the piston **450** can be achieved. In addition, since the inner seal **472** is interposed between the bottom surface **457** and the rear surface **132** over the entire circumference around the axial line L1, the place where the driving force acts between the plunger **100** and the piston **450** is dispersed over the entire circumference. As a result, the postures of both the plunger **100** and the piston **450** are stabilized, so that the plunger **100** and the piston can slide at higher-speed.

(124) The dispenser **1** may further comprise a through bolt **11** for attaching the guide block **200** to the cylinder **410**. The floatability of the piston **450** is more beneficial because the guide block **200** and the cylinder **410** are separate members from each other and a positional deviation of the

cylinder **410** relative to the guide block **200** is likely to occur.

(125) The dispenser **1** may further include a distal end surface **521** facing the proximal end surface **102** of the plunger **100** in the cylinder **410** to restrict the retraction of the plunger **100**; and a retraction limit adjustment portion **530** configured to adjust the position of the distal end surface **521** in a direction along the axial line **L1**. The distal end surface **521** acts directly on the cylinder **410**, and the most retracted position of the plunger **100** can be adjusted accurately.

(126) It is to be understood that not all aspects, advantages and features described herein may necessarily be achieved by, or included in, any one particular example. Indeed, having described and illustrated various examples herein, it should be apparent that other examples may be modified in arrangement and detail.

Claims

1. A dispenser comprising: a plunger comprising a distal end surface facing an advance direction of the plunger and a proximal end surface facing a retraction direction of the plunger; a dispensing portion configured to dispense and suction liquid in response to an advance and a retraction of the distal end surface along an axial line intersecting the distal end surface and the proximal end surface; wherein the dispensing portion comprises: a surrounding portion surrounding the distal end surface of the plunger about the axial line to define an accommodation chamber; and an end portion facing the distal end surface of the plunger along the axial line, wherein the end portion comprises an outlet opening into the accommodation chamber to dispense the liquid out of the accommodation chamber, wherein the surrounding portion comprises an inlet opening into the accommodation chamber to receive the liquid into the accommodation chamber, and wherein a distance from the inlet to the end portion is less than or equal to half of a maximum stroke of the distal end surface of the plunger.
2. The dispenser according to claim 1, wherein the dispensing portion further comprises: a seal portion facing the end portion so that the accommodation chamber is located between the seal portion and the end portion, and forming a seal between the surrounding portion and the plunger, and wherein the distance from the inlet to the end portion is less than a distance from the inlet to the seal portion.
3. The dispenser according to claim 1, wherein more than half of the maximum stroke of the distal end surface of the plunger overlaps the inlet along the axial line.
4. The dispenser according to claim 1, wherein the end portion further comprises: a dispense port opening outside of the accommodation chamber; and a dispense flow channel connecting the outlet and the dispense port without passing through a check valve.
5. The dispenser according to claim 1, further comprising: a pump configured to pump the liquid from outside the accommodation chamber to the inlet; and a pressure adjuster configured to adjust a pressure applied to the liquid by the pump.
6. The dispenser according to claim 5, wherein the pressure adjuster is configured to adjust the pressure applied to the liquid by the pump so that an inverse flow of the liquid from the accommodation chamber to the inlet is prevented during the advance of the plunger, and a dispense of the liquid from the outlet is temporarily ceased during the retraction of the plunger.
7. The dispenser according to claim 1, further comprising: a cylinder accommodating the plunger; a piston partitioning an inside of the cylinder into a first space and a second space, and configured to: advance the plunger toward the advance direction according to a pressure of the first space; and retract the plunger toward the retraction direction according to a pressure of the second space; and a piston driver configured to switch between a first state in which a pressure of a pressurizing source is applied to the first space and a second state in which the pressure of the pressurizing source is applied to the second space.
8. The dispenser according to claim 7, further comprising: a spring configured to apply a repulsive

force to the plunger toward the advance direction; and a release portion configured to release, by the pressure of the pressurizing source, an application of the repulsive force from the spring to the plunger.

9. The dispenser according to claim 8, further comprising a pusher configured to transmit the repulsive force from the spring to the plunger, wherein the release portion is configured to apply the pressure of the pressurizing source to the pusher toward the retraction direction to release the application of the repulsive force from the spring to the plunger.

10. The dispenser according to claim 9, wherein the release portion forms a third space in the cylinder, so that the pressure of the pressurizing source is applied to the third space in both the first state and the second state, and wherein the application of the repulsive force is released by the pressure of the third space.

11. The dispenser according to claim 10, wherein the first space is located between the second space and the third space along the axial line.

12. The dispenser according to claim 10, wherein the piston driver comprises: a pressurizing port pressurized by the pressurizing source; and a valve configured to switch between the first state in which the pressurizing port is connected to the first space and the second state in which the pressurizing port is connected to the second space, and wherein the third space is connected to the pressurizing port.

13. The dispenser according to claim 8, wherein the piston has an annular shape and partitions the inside of the cylinder into the first space and the second space between an outer peripheral surface of the plunger and an inner peripheral surface of the cylinder, and wherein the spring applies the repulsive force to the piston between the outer peripheral surface of the plunger and the inner peripheral surface of the cylinder.

14. The dispenser according to claim 7, wherein the piston has an annular shape and partitions the inside of the cylinder into the first space and the second space between an outer peripheral surface of the plunger and an inner peripheral surface of the cylinder, and wherein the dispenser further comprises a holder holding the piston on the outer peripheral surface of the plunger while allowing the piston to move in a direction vertical to the axial line.

15. The dispenser according to claim 14, further comprising a seal portion forming a seal between the piston and the plunger while allowing the piston to move in the direction vertical to the axial line.

16. The dispenser according to claim 15, wherein the piston has an annular first seal surface along a plane intersecting the axial line, wherein the plunger has an annular second seal surface facing the first seal surface, and the seal portion contacts the first seal surface and the second seal surface to form the seal between the piston and the plunger.

17. The dispenser according to claim 16, wherein the first seal surface faces the advance direction and the second seal surface faces the retraction direction.

18. The dispenser according to claim 17, wherein the plunger comprises a flange comprising the second seal surface and a restriction surface on an opposite side of the second seal surface, and wherein the holder holds the piston between the flange and the holder, and wherein the dispenser further comprises an advance restriction block configured to restrict the advance of the plunger by contacting the restriction surface.

19. The dispenser according to claim 14, further comprising: a guide portion configured to guide the advance and the retraction of the distal end surface along the axial line; and a fastening member for attaching the guide portion to the cylinder.

20. The dispenser according to claim 14, further comprising: a retraction restricting portion facing the proximal end surface of the plunger in the cylinder to restrict the retraction of the plunger; and a retraction limit adjustment portion configured to adjust a position of the retraction restricting portion in a direction along the axial line.
